

MINERvA with OmniFold

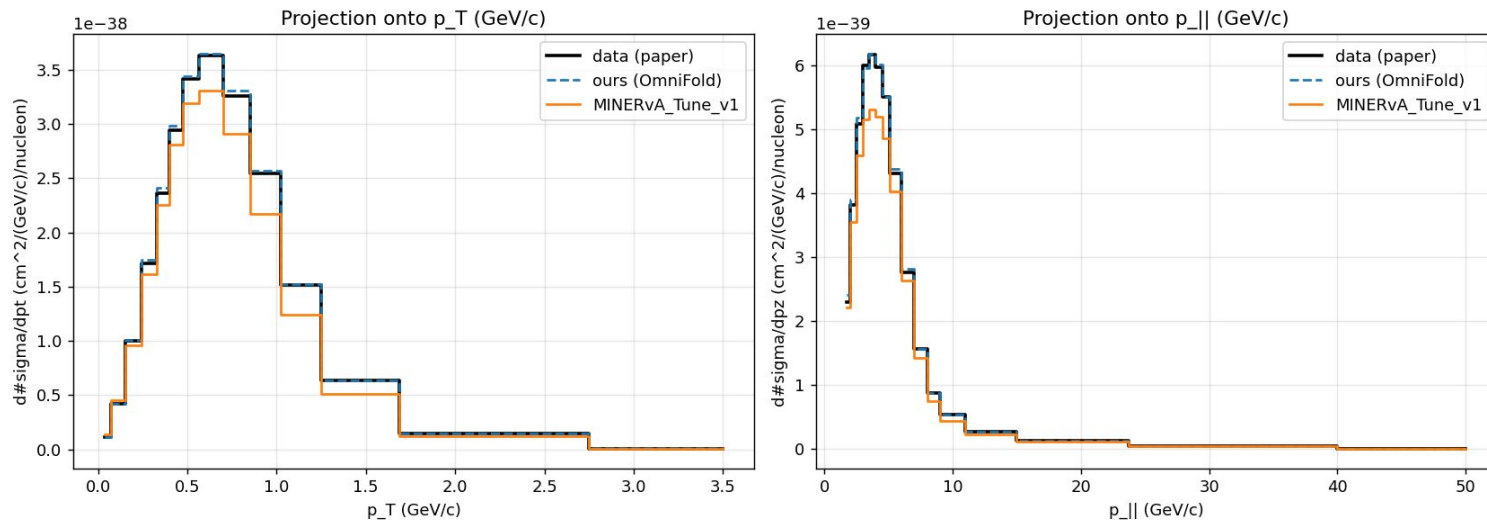
Generator Comparisons + Higher Dims
Last Updated: 6/5/26

Since last time:

Last time: 2D uncertainty quantification had a flux normalization error. OmniFold's combined median bin uncertainty is now 6.87% vs. their [6.86%](#).

In 2D the generators are consistent with the data shape (not true once we add E_{avail})

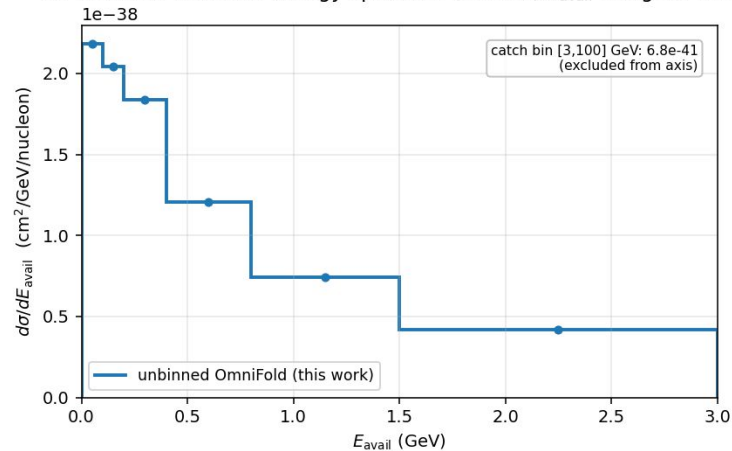
2D OmniFold vs data vs MINERvA Tune v1 - 1D projections



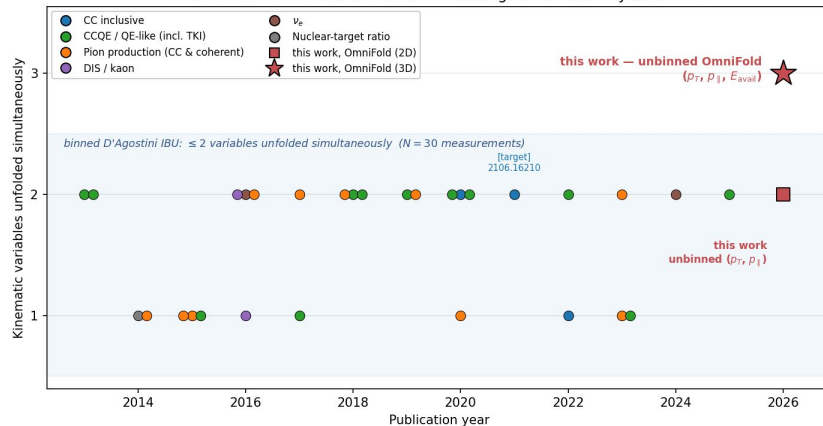
3D extension (+E_avail)

First 3D unfolding with MINERvA!

CC-inclusive available-energy spectrum (MEFHC, E_{avail} -marginal of 3D



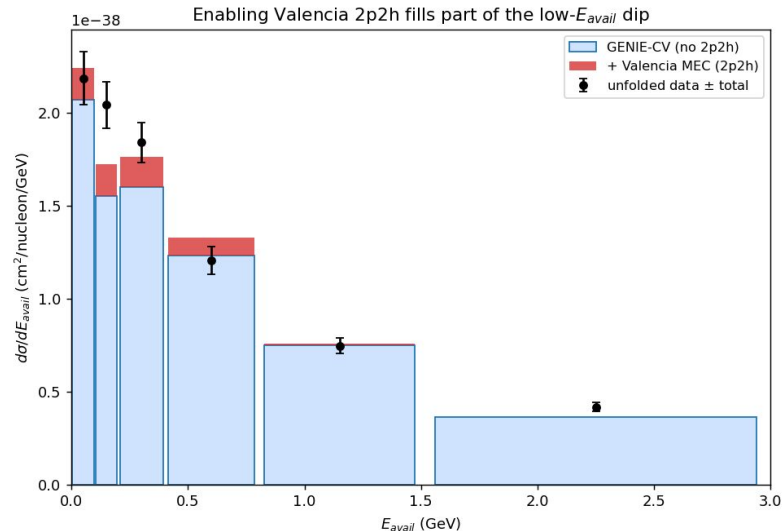
MINERvA differential cross-section unfolding: dimensionality over time



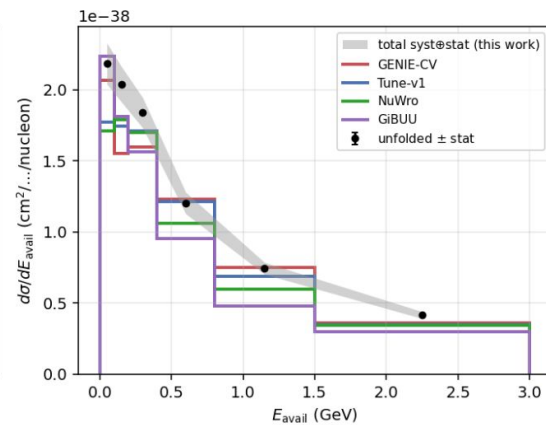
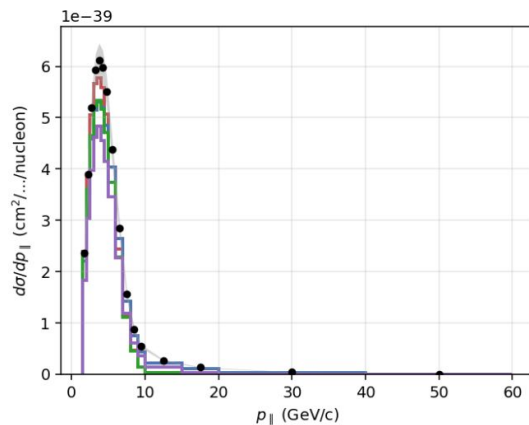
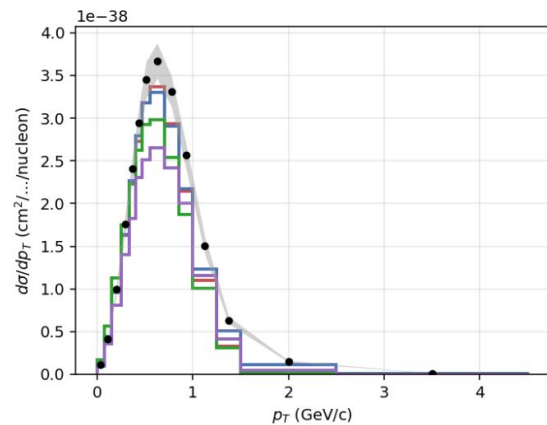
3D Physics

Low-recoil excess, 4 generators under-predict at low E_{avail}

Looks like missing 2p2h

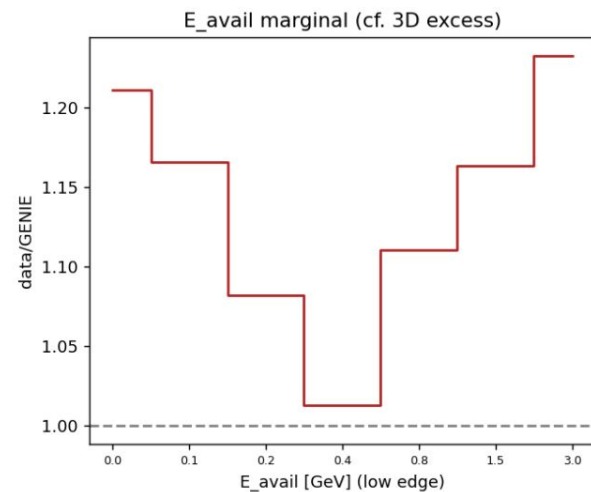
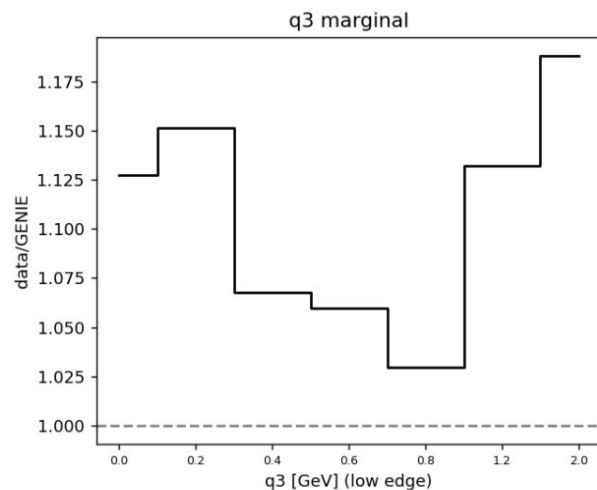
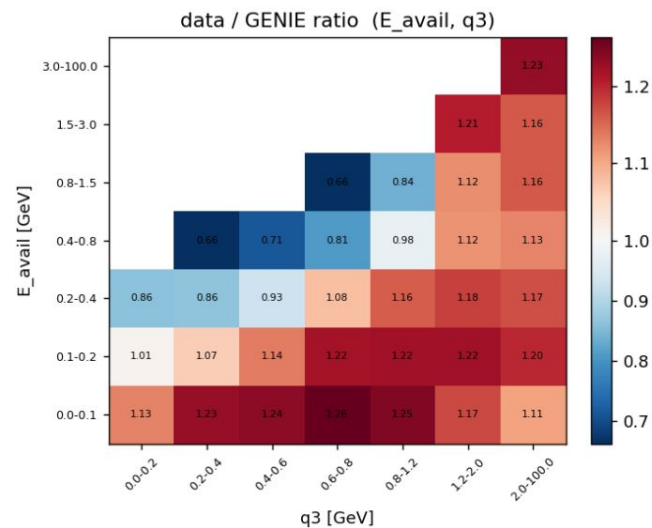


Unfolded 3D result ($\pm$ total syst@stat) vs generators (Eavail catch bin omitted)



4D extension (+q3)

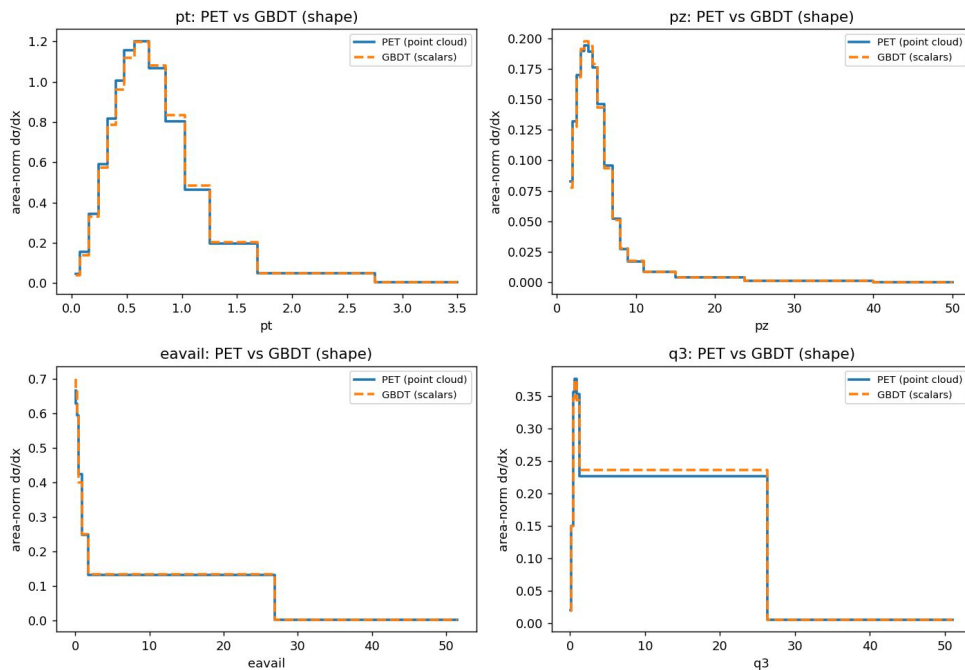
E_{avail} -marginal reproduces 3D (closure); (E_{avail}, q_3) has real structure; covariance is model-dominated.



Method validation

Unbinned GoF passes; PET point-cloud approach (from ViniciusMikuni/omnifold) reproduces GBDT.

Phase-3: does the point cloud change the unfolded shape vs scalars?



Status

Computing stats+ML uncertainties for 4D.

Need Ascencio low- q_3 data release (arXiv:2110.13372 / PRD 106 032001) in order to make a comparison like we did for 2D. They do a $d^2\sigma/(dq_3 dE_{\text{avail}})$ unfolding here.

+ 2.2σ high- E_{avail} DIS-tail, should we look into it?